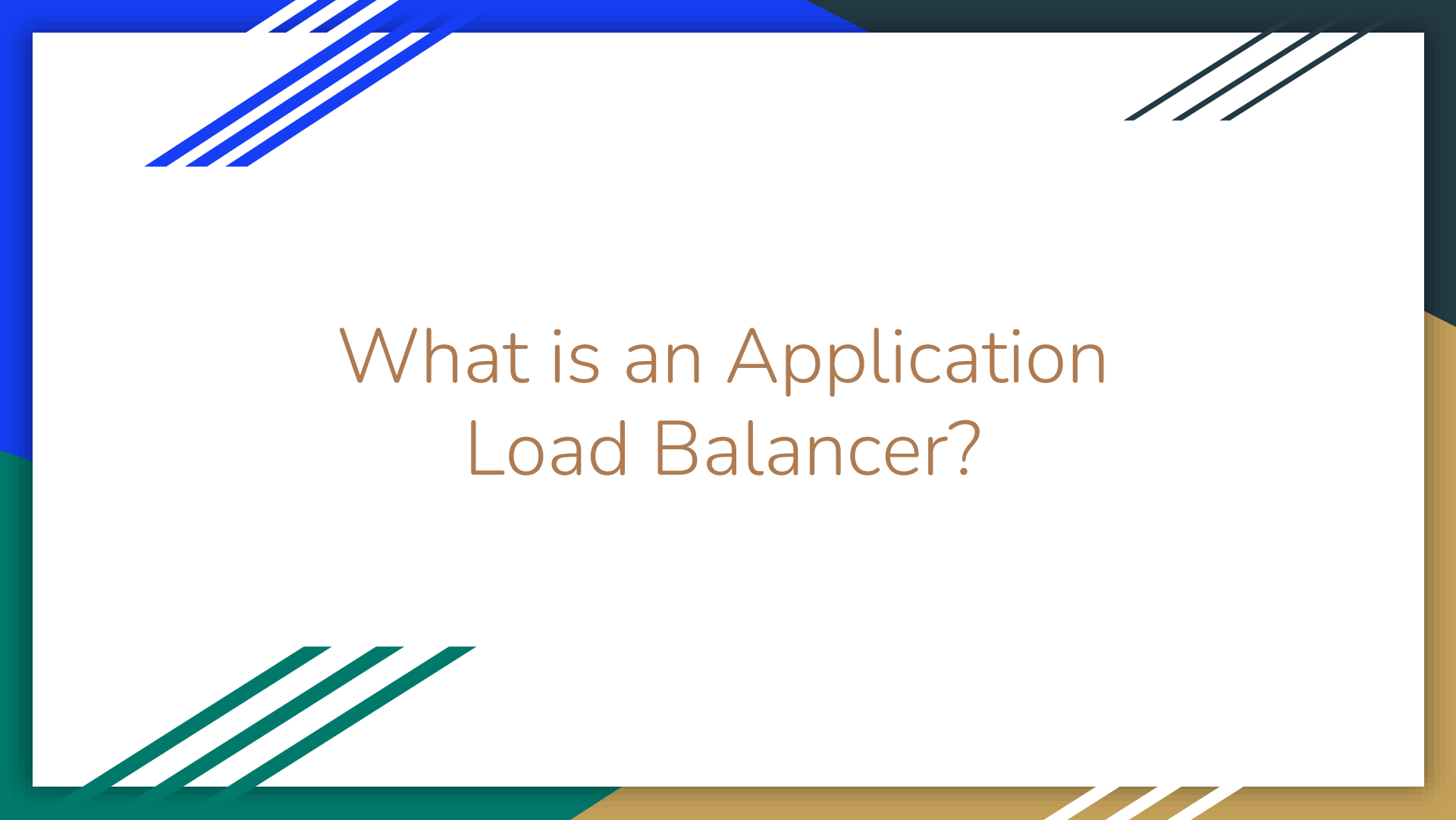




What is an Application
Load Balancer?

What is an Application Load Balancer?

Elastic Load Balancing automatically distributes your incoming traffic across multiple targets, such as EC2 instances, containers, and IP addresses, in one or more Availability Zones. It monitors the health of its registered targets, and routes traffic only to the healthy targets. Elastic Load Balancing scales your load balancer as your incoming traffic changes over time. It can automatically scale to the vast majority of workloads.

Elastic Load Balancing supports the following load balancers: Application Load Balancers, Network Load Balancers, Gateway Load Balancers, and Classic Load Balancers. You can select the type of load balancer that best suits your needs. This guide discusses Application Load Balancers.

Application Load Balancer components

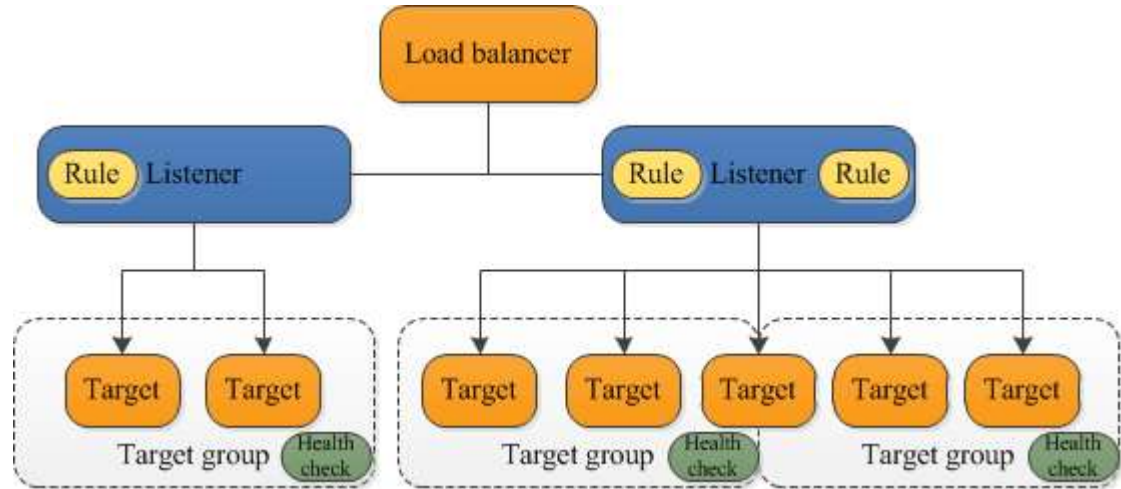
A load balancer serves as the single point of contact for clients. The load balancer distributes incoming application traffic across multiple targets, such as EC2 instances, in multiple Availability Zones. This increases the availability of your application. You add one or more listeners to your load balancer.

A listener checks for connection requests from clients, using the protocol and port that you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets. Each rule consists of a priority, one or more actions, and one or more conditions. When the conditions for a rule are met, then its actions are performed. You must define a default rule for each listener, and you can optionally define additional rules.

Application Load Balancer components

Each target group routes requests to one or more registered targets, such as EC2 instances, using the protocol and port number that you specify. You can register a target with multiple target groups. You can configure health checks on a per target group basis. Health checks are performed on all targets registered to a target group that is specified in a listener rule for your load balancer.

The following diagram illustrates the basic components. Notice that each listener contains a default rule, and one listener contains another rule that routes requests to a different target group. One target is registered with two target groups.



Application Load Balancer overview

An Application Load Balancer functions at the application layer, the seventh layer of the Open Systems Interconnection (OSI) model. After the load balancer receives a request, it evaluates the listener rules in priority order to determine which rule to apply, and then selects a target from the target group for the rule action. You can configure listener rules to route requests to different target groups based on the content of the application traffic. Routing is performed independently for each target group, even when a target is registered with multiple target groups. You can configure the routing algorithm used at the target group level. The default routing algorithm is round robin; alternatively, you can specify the least outstanding requests routing algorithm.

You can add and remove targets from your load balancer as your needs change, without disrupting the overall flow of requests to your application. Elastic Load Balancing scales your load balancer as traffic to your application changes over time. Elastic Load Balancing can scale to the vast majority of workloads automatically.

You can configure health checks, which are used to monitor the health of the registered targets so that the load balancer can send requests only to the healthy targets.

Benefits of migrating from a Classic Load Balancer

Using an Application Load Balancer instead of a Classic Load Balancer has the following benefits:

- Support for Path conditions. You can configure rules for your listener that forward requests based on the URL in the request. This enables you to structure your application as smaller services, and route requests to the correct service based on the content of the URL.
- Support for Host conditions. You can configure rules for your listener that forward requests based on the host field in the HTTP header. This enables you to route requests to multiple domains using a single load balancer.
- Support for routing based on fields in the request, such as HTTP header conditions and methods, query parameters, and source IP addresses.
- Support for routing requests to multiple applications on a single EC2 instance. You can register each instance or IP address with the same target group using multiple ports.

Benefits of migrating from a Classic Load Balancer

- Support for redirecting requests from one URL to another.
- Support for returning a custom HTTP response.
- Support for registering targets by IP address, including targets outside the VPC for the load balancer.
- Support for registering Lambda functions as targets.
- Support for the load balancer to authenticate users of your applications through their corporate or social identities before routing requests.
- Support for containerized applications. Amazon Elastic Container Service (Amazon ECS) can select an unused port when scheduling a task and register the task with a target group using this port. This enables you to make efficient use of your clusters.

Benefits of migrating from a Classic Load Balancer

- Support for monitoring the health of each service independently, as health checks are defined at the target group level and many CloudWatch metrics are reported at the target group level. Attaching a target group to an Auto Scaling group enables you to scale each service dynamically based on demand.
- Access logs contain additional information and are stored in compressed format.
- Improved load balancer performance.



Getting started with Application Load Balancers



Step 1: Select a load balancer type

Elastic Load Balancing supports different types of load balancers. For this tutorial, you create an Application Load Balancer.

To create an Application Load Balancer

1. Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>.
2. On the navigation bar, choose a region for your load balancer. Be sure to select the same region that you used for your EC2 instances.
3. On the navigation pane, under **LOAD BALANCING**, choose **Load Balancers**.
4. Choose **Create Load Balancer**.
5. For **Application Load Balancer**, choose **Create**.

Step 2: Configure your load balancer and listener

On the Configure Load Balancer page, complete the following procedure.

To configure your load balancer and listener

1. For **Name**, enter a name for your load balancer.
The name of your Application Load Balancer must be unique within your set of Application Load Balancers and Network Load Balancers for the region, can have a maximum of 32 characters, can contain only alphanumeric characters and hyphens, must not begin or end with a hyphen, and must not begin with "internal-".
2. For **Scheme** and **IP address** type, keep the default values.
3. For **Listeners**, keep the default, which is a listener that accepts HTTP traffic on port 80.
4. For **Availability Zones**, select the VPC that you used for your EC2 instances. For each Availability Zone that you used to launch your EC2 instances, select the Availability Zone and then select the public subnet for that Availability Zone.
5. Choose Next: **Configure Security Settings**.
6. For this tutorial, you are not creating an HTTPS listener. Choose **Next: Configure Security Groups**.

Step 3: Configure a security group for your load balancer

The security group for your load balancer must allow it to communicate with registered targets on both the listener port and the health check port. The console can create a security group for your load balancer on your behalf, with rules that specify the correct protocols and ports. If you prefer, you can create and select your own security group instead.

On the **Configure Security Groups** page, complete the following procedure to have Elastic Load Balancing create a security group for your load balancer on your behalf.

To configure a security group for your load balancer

1. Choose Create a new security group.
2. Type a name and description for the security group, or keep the default name and description. This new security group contains a rule that allows traffic to the load balancer listener port that you selected on the **Configure Load Balancer** page.
3. Choose **Next: Configure Routing**.

Step 4: Configure your target group

Create a target group, which is used in request routing. The default rule for your listener routes requests to the registered targets in this target group. The load balancer checks the health of targets in this target group using the health check settings defined for the target group. On the **Configure Routing** page, complete the following procedure.

To configure your target group

1. For **Target group**, keep the default, **New target group**.
2. For **Name**, enter a name for the new target group.
3. Keep the default target type (**Instance**), protocol (**HTTP**), and port (**80**).
4. For **Health checks**, keep the default settings.
5. Choose **Next: Register Targets**.

Step 5: Register targets with your target group

On the **Register Targets** page, complete the following procedure.

To register your instances with the target group

1. For **Instances**, select one or more instances.
2. Keep the default port (**80**) and choose **Add to registered**.
3. When you have finished selecting instances, choose **Next: Review**.

Step 6: Create and test your load balancer

Before creating the load balancer, review the settings that you selected. After creating the load balancer, verify that it's sending traffic to your EC2 instances.

To create and test your load balancer

1. On the **Review** page, choose **Create**.
2. After you are notified that your load balancer was created successfully, choose **Close**.
3. On the navigation pane, under **LOAD BALANCING**, choose **Target Groups**.
4. Select the newly created target group.
5. On the **Targets** tab, verify that your instances are ready. If the status of an instance is `initial`, it's probably because the instance is still in the process of being registered, or it has not passed the minimum number of health checks to be considered healthy. After the status of at least one instance is `healthy`, you can test your load balancer.

Step 6: Create and test your load balancer

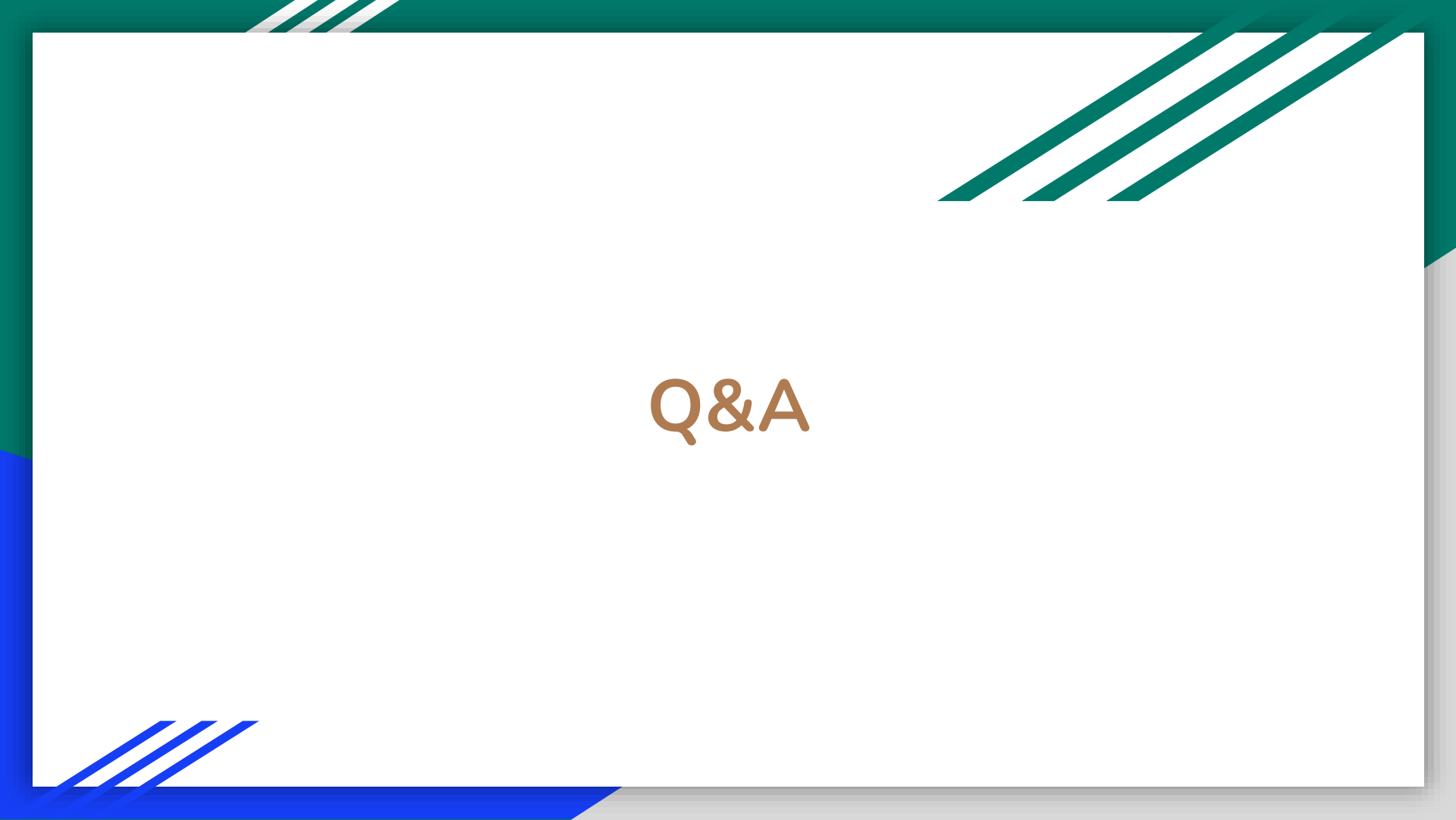
6. On the navigation pane, under **LOAD BALANCING**, choose **Load Balancers**.
7. Select the newly created load balancer.
8. On the **Description** tab, copy the DNS name of the load balancer (for example, my-load-balancer-1234567890.us-west-2.elb.amazonaws.com). Paste the DNS name into the address field of an Internet-connected web browser. If everything is working, the browser displays the default page of your server.
9. (Optional) To define additional listener rules, see Add a rule.

Step 7: Delete your load balancer (optional)

As soon as your load balancer becomes available, you are billed for each hour or partial hour that you keep it running. When you no longer need a load balancer, you can delete it. As soon as the load balancer is deleted, you stop incurring charges for it. Note that deleting a load balancer does not affect the targets registered with the load balancer. For example, your EC2 instances continue to run.

To delete your load balancer

1. Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>.
2. On the navigation pane, under **LOAD BALANCING**, choose **Load Balancers**.
3. Select the checkbox for the load balancer, and then choose **Actions, Delete**.
4. When prompted for confirmation, choose **Yes, Delete**.



Q&A